

Dimensionality Reduction

Complete Exam Revision Guide

Covers: PCA & Feature Selection

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1 Introduction to Dimensionality Reduction

1.1 Definition & Core Concepts

Dimensionality Reduction: A technique that simplifies data by reducing the number of variables/dimensions while preserving essential information.

1.1.1 Key Principles

- **Preserving Relevance:** Retains crucial information from high-dimensional spaces
- **Eliminating Redundancy:** Removes redundant and irrelevant attributes
- **Transformation:** Maps data into a lower-dimensional, manageable space

1.2 Motivations for Dimensionality Reduction

1.2.1 1. Improve Model Performance

- *Better accuracy:* Simpler models often generalize better
- *Reduce overfitting:* Fewer features = fewer parameters to fit
- *Example:* A model with 100 features might overfit on small datasets, but 10 key features can capture the essential patterns

1.2.2 2. Efficiency & Resource Management

- *Computational efficiency:* Faster training and prediction
- *Memory savings:* Less storage required
- *Real-world impact:* Mobile apps and embedded systems benefit greatly

1.2.3 3. Interpretability & Understanding

- *Visualization:* Can plot 2D/3D representations of complex data
- *Enhanced interpretability:* Easier to understand which factors matter
- *Communication:* Stakeholders can grasp insights more easily

1.2.4 4. Data Quality Enhancement

- *Noise reduction:* Eliminates noisy, irrelevant features
- *Focus on signal:* Keeps informative attributes

1.3 The Curse of Dimensionality

1.3.1 Core Problem

As dimensions increase, data becomes increasingly sparse in the feature space.

Key Mathematical Insight:

$$\frac{V_{\text{sphere}}}{V_{\text{cube}}} = \frac{\pi^{d/2}}{2^d \Gamma(d/2 + 1)} \rightarrow 0 \text{ as } d \rightarrow \infty \quad (1)$$

What this means: In a d -dimensional unit hypercube, almost all volume is in the corners, not near the center.

1.3.2 Consequences

1. **Sparsity:** Data points become far apart from each other and the center
2. **Distance metrics fail:** Euclidean distance loses meaning
 - In high dimensions: $d_{\max} - d_{\min} \rightarrow 0$
 - All points appear equidistant!
3. **Algorithm failures:** Nearest-neighbor, clustering, classification struggle
4. **Sample size requirements:** Need exponentially more data

Exam Tip: Be able to explain why distance-based algorithms fail in high dimensions.

2 Two Main Approaches

2.1 Feature Extraction vs. Feature Selection

Feature Extraction	Feature Selection
Transforms original features into new ones	Selects subset of original features
Example: PCA creates new axes	Example: Keep features A, C, E; discard B, D
New features are combinations	Original features preserved
Loses original measurement units	Keeps original measurement units
Always numeric output	Can handle non-numeric features
Linear (PCA) or Non-linear (t-SNE)	Filter, Wrapper, or Embedded methods

3 Principal Component Analysis (PCA)

3.1 Motivation Example: Spring Motion

Scenario: Track a ball on an oscillating spring with 3 cameras

- Each camera records (x, y) position
- Data: 6D vector $X = [x_A, y_A, x_B, y_B, x_C, y_C]$
- *Physical reality:* Motion is 1D along the spring
- *Question:* How to extract this 1D dynamic from 6D recordings?

Answer: PCA identifies that one direction (principal component) captures most variation.

3.2 What is PCA?

3.2.1 Goal

Transform original data into new uncorrelated variables that align with data variation.

3.2.2 Principle

Rotate axes to align with directions of maximum variability in the data.

3.2.3 Principal Components

- **PC1 (First Principal Component):** Direction of maximum variance
- **PC2:** Orthogonal to PC1, second-most variance
- **PC k :** Orthogonal to all previous, k -th most variance
- Each PC is a linear combination of original features

3.3 Two Mathematical Formulations

3.3.1 1. Maximizing Variance (Hotelling 1933)

Find directions where data varies the most.

$$\max_{\mathbf{w}} \text{Var}(\mathbf{w}^T \mathbf{X}) \quad \text{subject to} \quad \|\mathbf{w}\| = 1 \quad (2)$$

3.3.2 2. Minimizing Projection Error (Pearson 1901)

Find directions that minimize reconstruction error.

$$\min_{\mathbf{w}} \sum_{i=1}^n \|\mathbf{x}_i - \mathbf{w}\mathbf{w}^T \mathbf{x}_i\|^2 \quad (3)$$

Key Insight: These two formulations are equivalent!

- Maximizing remaining variance $\Leftrightarrow$ Minimizing projection error
- Total variance = Remaining variance + Lost variance (constant)

3.4 PCA Algorithm - Step by Step

Input: Data matrix $\mathbf{X}$ of size $n \times m$ (n samples, m features)

3.4.1 Step 1: Preprocessing

a) **Centering (Mandatory):**

$$\mathbf{X}_{\text{centered}} = \mathbf{X} - \boldsymbol{\mu} \quad (4)$$

where $\boldsymbol{\mu}$ is the mean of each column.

Why? Makes covariance calculation simpler and centers PC origin at data mean.

b) **Scaling (Optional but Recommended):**

$$\mathbf{X}_{\text{scaled}} = \frac{\mathbf{X} - \boldsymbol{\mu}}{\sigma} \quad (5)$$

where σ is standard deviation of each column.

When to scale?

- *YES:* Features have different units (e.g., height in cm, weight in kg)
- *NO:* All features in same units and scale matters (e.g., all in dollars)

3.4.2 Step 2: Covariance Matrix

$$\boldsymbol{\Sigma} = \frac{1}{n-1} \mathbf{X}^T \mathbf{X} \quad (6)$$

Result: $m \times m$ symmetric matrix where:

- **Diagonal:** Variance of each feature
- **Off-diagonal:** Covariance between feature pairs

Exam Note: If data is centered, $\boldsymbol{\Sigma} = \frac{1}{n-1} \mathbf{X}^T \mathbf{X}$ simplifies computation.

3.4.3 Step 3: Eigenvalue Decomposition

Diagonalize the covariance matrix:

$$\boldsymbol{\Sigma} = \mathbf{P} \mathbf{D} \mathbf{P}^T \quad (7)$$

where:

- **D:** Diagonal matrix of eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_m$
- **P:** Matrix of corresponding eigenvectors (columns)
- Eigenvectors are orthogonal (perpendicular to each other)

Alternative: Can use Singular Value Decomposition (SVD) for numerical stability.

3.4.4 Step 4: Select Top k Components

Sort eigenvalues descending and choose top k eigenvectors:

- **$\mathbf{P}_k$:** $m \times k$ matrix containing top k eigenvectors
- These become the principal components

3.4.5 Step 5: Project Data

Transform data to new coordinate system:

$$\mathbf{X}_{\text{PCA}} = \mathbf{X} \cdot \mathbf{P}_k \quad (8)$$

Result: $n \times k$ matrix with data in reduced k -dimensional space.

3.4.6 Reconstruction (Optional)

Approximate original data from reduced representation:

$$\mathbf{X}_{\text{approx}} = \mathbf{X}_{\text{PCA}} \cdot \mathbf{P}_k^T \quad (9)$$

Note: $\mathbf{X}_{\text{approx}} \approx \mathbf{X}$ with some loss (the projection error).

3.5 Choosing the Number of Components k

3.5.1 Method 1: Variance Retention Threshold

Choose smallest k such that:

$$\frac{\sum_{i=1}^k \lambda_i}{\sum_{i=1}^m \lambda_i} \geq 0.95 \text{ (or } 0.99) \quad (10)$$

Interpretation: Retain 95% (or 99%) of total variance.

3.5.2 Method 2: Scree Plot

- Plot eigenvalues vs. component number
- Look for "elbow" - sharp drop-off point
- Choose k at the elbow

Exam Tip: Be able to identify elbow in a scree plot diagram.

3.5.3 Method 3: Cross-Validation

Try different k values and evaluate downstream task performance.

3.6 Applications of PCA

3.6.1 1. Data Compression

- Reduce storage requirements
- Example: Images (JPEG uses similar DCT transform)

3.6.2 2. Denoising

- High-variance components = signal
- Low-variance components = noise
- Keep only high-variance PCs

3.6.3 3. Visualization

- Project to 2D or 3D for plotting
- Explore cluster structure
- Identify patterns and outliers

3.6.4 4. Face Recognition (Eigenfaces)

- Each face image $\rightarrow$ high-dimensional vector
- PCA finds principal "eigenfaces"
- New faces represented as combinations
- Compare coefficients for recognition

3.6.5 5. Other Applications

- Anomaly detection
- Recommendation systems
- Genomic data analysis
- Financial portfolio optimization

3.7 PCA Summary - Key Points for Exam

1. PCA finds orthogonal directions of maximum variance
2. Always center data; scale if features have different units
3. Eigenvalues indicate importance (variance) of each component
4. Eigenvectors define the new coordinate system
5. Choose k based on variance retention or scree plot
6. PCA is *linear* - cannot capture non-linear relationships
7. PCA is *unsupervised* - doesn't use class labels

4 Feature Selection

4.1 Why Feature Selection?

4.1.1 When to Use Feature Selection vs. Feature Extraction?

Use Feature Selection when:

1. Features are expensive to collect (e.g., medical tests, sensors)
2. Need interpretable rules (domain experts must understand)
3. Features are non-numeric (categorical data)
4. Want to identify truly important original features
5. Need to preserve original measurement units

Use Feature Extraction (PCA) when:

1. All features are numeric
2. Don't need to interpret individual features
3. Features are correlated (PCA removes correlation)
4. Want maximum information retention

4.2 The Combinatorial Challenge

For n features, number of possible subsets:

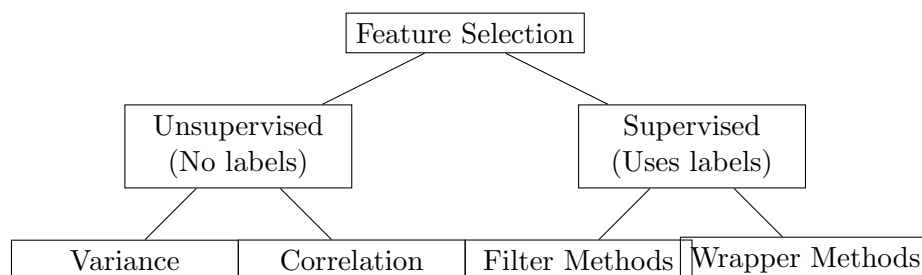
$$2^n - 1 \text{ subsets} \quad (11)$$

Example:

- $n = 3$: 7 subsets (manageable)
- $n = 10$: 1,023 subsets
- $n = 20$: 1,048,575 subsets
- $n = 100$: 1.27×10^{30} subsets (impossible!)

Solution: Use heuristic search strategies (not exhaustive search).

4.3 Taxonomy of Feature Selection



5 Unsupervised Feature Selection

Uses statistical measures without considering target variable.

5.1 1. Variance Filter

Principle: Remove features with low variance (nearly constant values).

Formula:

$$\text{Var}(X) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \quad (12)$$

Example:

- Feature A: [1, 1, 1, 1, 1] → Var = 0 (remove!)
- Feature B: [1, 5, 2, 8, 3] → Var = 7.2 (keep!)

Use case: First step to remove obviously useless features.

5.2 2. Correlation Filter

Principle: Remove highly correlated features (redundant information).

Pearson Correlation:

$$r_{XY} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \quad (13)$$

Range: $-1 \leq r \leq 1$

Decision rule:

- If $|r_{XY}| > 0.9$: Features X and Y are redundant, remove one
- Keep features with low correlation

Limitation: Only detects *linear* relationships.

5.3 3. Mutual Information

Principle: Measure dependency between categorical features (detects any relationship).

Formula:

$$I(X; Y) = H(X) + H(Y) - H(X, Y) \quad (14)$$

where $H(X)$ is entropy:

$$H(X) = - \sum_i p(x_i) \log_2 p(x_i) \quad (15)$$

Properties:

- $I(X; Y) = 0$: Independent features
- $I(X; Y) > 0$: Dependent features
- Can detect non-linear relationships (unlike correlation)

6 Supervised Feature Selection

Uses target variable to evaluate features.

6.1 Filter Methods

Goal: Rank features independently using statistical measures with target.

Process:

1. Choose metric (correlation, information gain, chi-square)
2. Calculate metric for each feature vs. target
3. Rank features by metric score
4. Select top k features

6.1.1 1. Correlation with Target

For continuous target: Use Pearson correlation

- High $|r|$: Feature strongly related to target (keep)
- Low $|r|$: Feature weakly related to target (discard)

Limitation: Misses non-linear relationships and feature interactions.

Exam Example:

- Target: House price
- Feature 1: Square footage, $r = 0.85$ (strong, keep)
- Feature 2: Owner's favorite color, $r = 0.02$ (weak, discard)

6.1.2 2. Information Gain

For categorical target and features: Measure entropy reduction.

Formula:

$$IG(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v) \quad (16)$$

where:

- S : Entire dataset
- A : Feature to evaluate
- S_v : Subset where feature A has value v
- $H(S)$: Entropy of target in set S

Entropy formula:

$$H(S) = - \sum_{i=1}^c p_i \log_2(p_i) \quad (17)$$

where p_i is proportion of class i .

Interpretation:

- $IG = 0$: Feature provides no information
- $IG = 1$: Feature perfectly predicts target

Age	Road Tested	Bought
Recent	Yes	Yes
Recent	Yes	Yes
Recent	No	Yes
Recent	No	No
Old	Yes	Yes
Old	Yes	Yes
Old	No	No
Old	No	No

- Higher IG $\rightarrow$ More useful feature

Worked Example:

Dataset: Predict if a car will be bought (Yes/No)

Step 1: Calculate root entropy

- $p(\text{Yes}) = 5/8 = 0.625$
- $p(\text{No}) = 3/8 = 0.375$
- $H(\text{Root}) = -(0.625 \log_2(0.625) + 0.375 \log_2(0.375)) = 0.954$

Step 2: Calculate IG for "Age"

- Recent: 3 Yes, 1 No $\rightarrow H = 0.811$
- Old: 2 Yes, 2 No $\rightarrow H = 1.0$
- $IG(\text{Age}) = 0.954 - (4/8 \times 0.811 + 4/8 \times 1.0) = 0.048$

Step 3: Calculate IG for "Road Tested"

- Yes: 4 Yes, 0 No $\rightarrow H = 0$ (pure!)
- No: 1 Yes, 3 No $\rightarrow H = 0.811$
- $IG(\text{Road Tested}) = 0.954 - (4/8 \times 0 + 4/8 \times 0.811) = 0.548$

Conclusion: "Road Tested" is more informative than "Age".

6.1.3 3. Chi-Square Test

For categorical features and target: Test independence.

Hypothesis:

- H_0 : Feature and target are independent (not related)
- H_1 : Feature and target are dependent (related)

Formula:

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \quad (18)$$

where:

- O_{ij} : Observed frequency in cell (i, j)
- E_{ij} : Expected frequency if independent

Expected frequency:

$$E_{ij} = \frac{(\text{Row}_i \text{ total}) \times (\text{Column}_j \text{ total})}{\text{Grand total}} \quad (19)$$

Decision rule:

1. Calculate χ^2 statistic
2. Determine degrees of freedom: $df = (r - 1)(c - 1)$
3. Look up critical value from chi-square table at significance level α (usually 0.05)
4. If $\chi^2 > \chi^2_{\text{critical}}$: Reject H_0 (feature is related, keep it!)
5. If $\chi^2 \leq \chi^2_{\text{critical}}$: Accept H_0 (feature independent, discard)

Exam Example:

	Cat Lover	Dog Lover	Total
Male	18	32	50
Female	27	23	50
Total	45	55	100

Expected values:

- $E_{\text{Male,Cat}} = (50 \times 45)/100 = 22.5$
- $E_{\text{Male,Dog}} = (50 \times 55)/100 = 27.5$
- $E_{\text{Female,Cat}} = 22.5$
- $E_{\text{Female,Dog}} = 27.5$

Chi-square:

$$\chi^2 = \frac{(18 - 22.5)^2}{22.5} + \frac{(32 - 27.5)^2}{27.5} \quad (20)$$

$$+ \frac{(27 - 22.5)^2}{22.5} + \frac{(23 - 27.5)^2}{27.5} \quad (21)$$

$$= 0.9 + 0.736 + 0.9 + 0.736 = 3.27 \quad (22)$$

Conclusion:

- $df = (2 - 1)(2 - 1) = 1$
- $\chi^2_{\text{critical}}(df = 1, \alpha = 0.05) = 3.84$
- $3.27 < 3.84 \rightarrow$ Cannot reject H_0
- Gender and pet preference appear independent (though close!)

6.2 Wrapper Methods

Goal: Evaluate feature subsets using actual model performance.

Key Difference from Filters:

- Filters: Evaluate features *independently*
- Wrappers: Evaluate feature *subsets* using model

6.2.1 General Wrapper Process

1. **Search Strategy:** Generate candidate feature subsets
2. **Model Training:** Train model on each subset
3. **Evaluation:** Measure model performance (accuracy, F1, etc.)
4. **Selection:** Choose subset with best performance
5. **Iteration:** Repeat until stopping criterion met

Stopping Criteria:

- Reach desired number of features
- Performance threshold achieved
- No improvement in recent iterations
- Elbow in performance curve

6.2.2 Search Strategies

1. Forward Selection (Bottom-up)

1. Start: Empty feature set $S = \{\}$
2. Loop:
 - Try adding each remaining feature
 - Evaluate model performance with each addition
 - Add feature that gives best improvement
3. Stop: When desired k features or no improvement

Complexity: $O(m^2)$ where m is number of features

Advantage: Fast, works well when few features are relevant

Disadvantage: May miss feature interactions (can't remove once added)

2. Backward Elimination (Top-down)

1. Start: All features $S = \{A_1, A_2, \dots, A_m\}$
2. Loop:
 - Try removing each current feature
 - Evaluate model performance with each removal
 - Remove feature whose removal hurts least (or helps)
3. Stop: When desired k features or performance drops

Complexity: $O(m^2)$

Advantage: Considers all features initially, good for feature interactions

Disadvantage: Expensive to start, requires training on all features

3. Genetic Algorithm (Evolutionary)

Encoding: Binary string where 1 = feature selected, 0 = feature discarded

Example: [1, 0, 1, 1, 0] means select features 1, 3, 4

Algorithm:

1. **Initialize:** Random population of feature subsets
2. **Evaluate:** Train model and measure fitness (accuracy) for each
3. **Select:** Choose best-performing subsets (parents)
4. **Crossover:** Combine parent solutions to create offspring
 - Example: Parent1 = [1,0,1,1,0], Parent2 = [0,1,1,0,1]
 - Offspring = [1,0,1,0,1] (take first half from P1, second from P2)
5. **Mutation:** Randomly flip bits with small probability
 - Example: [1,0,1,0,1] $\rightarrow$ [1,1,1,0,1] (flip position 2)
6. **Iterate:** Repeat for many generations
7. **Output:** Best solution found

Advantages:

- Explores diverse feature combinations
- Can escape local optima
- Handles feature interactions well

Disadvantages:

- Computationally expensive
- Many hyperparameters to tune
- Non-deterministic results

6.3 Filter vs. Wrapper Comparison

Aspect	Filter Methods	Wrapper Methods
Speed	Fast	Slow
Model dependency	Independent	Dependent on specific model
Feature interactions	Ignores	Captures
Optimality	Suboptimal	Can be optimal for given model
Scalability	High (large datasets)	Low (small-medium datasets)
Interpretability	Easy to interpret scores	Black box
Overfitting risk	Lower	Higher (especially with small data)
Use case	Initial screening, large m	Fine-tuning, small-medium m

Exam Tip: Be able to recommend which method based on scenario!

7 Comparison Summary: All Methods

Method	Supervised?	Speed	Feature Type	Best Use Case
Variance	No	Very Fast	Numeric	Remove constant features
Correlation	No	Fast	Numeric	Remove redundant features
Mutual Info	No	Fast	Categorical	Detect any relationship
Filter (Corr)	Yes	Fast	Numeric	Large datasets, linear relationships
Filter (Info Gain)	Yes	Fast	Categorical	Classification with categorical features
Filter (Chi-sq)	Yes	Fast	Categorical	Test independence
Wrapper (Forward)	Yes	Medium	Any	When few features expected to be relevant
Wrapper (Backward)	Yes	Slow	Any	When most features expected to be relevant
Wrapper (GA)	Yes	Very Slow	Any	Complex feature interactions, small m
PCA	No	Medium	Numeric	Correlated features, visualization, compression

8 Decision Framework: Which Method to Choose?

8.1 Consider These Factors

8.1.1 1. Dataset Size

- **Large ($n > 10,000$):** Use filter methods
- **Small ($n < 1,000$):** Can use wrapper methods
- **High dimensions ($m > 100$):** Start with filters, then wrappers

8.1.2 2. Computational Budget

- **Limited time:** Filters only
- **Moderate time:** Filter + Forward selection
- **Unlimited time:** Filter + Wrapper (GA or backward)

8.1.3 3. Interpretability Needs

- **Need to explain:** Feature selection (keep original features)
- **Just need performance:** PCA or any method
- **Domain expert review:** Filter methods (show feature scores)

8.1.4 4. Feature Types

- **All numeric:** PCA or any method
- **All categorical:** Information Gain, Chi-square, Wrappers
- **Mixed:** Wrappers or encode then use any

8.1.5 5. Label Availability

- **No labels (unsupervised):** Variance, Correlation, Mutual Info, PCA
- **Have labels (supervised):** Filter or Wrapper with target

8.1.6 6. Algorithm Specificity

- **Algorithm-independent:** Filter methods, PCA
- **Optimize for specific algorithm:** Wrapper methods

8.1.7 7. Feature Interactions

- **Features independent:** Filter methods work well
- **Strong interactions expected:** Wrapper methods, PCA

8.2 Recommended Strategies

Strategy 1: Two-Stage Approach (Most Common)

1. Stage 1: Use variance + correlation filter (remove obviously bad features)
2. Stage 2: Use wrapper method on remaining features

Strategy 2: Hybrid Approach

1. Run multiple filter methods (correlation, info gain, chi-square)
2. Take union of top features from each
3. Use wrapper to fine-tune

Strategy 3: PCA + Selection

1. Use PCA to understand feature correlations
2. Examine principal component loadings
3. Select original features with high loadings

9 Worked Examples for Exam Practice

9.1 Example 1: Choosing Between PCA and Feature Selection

Scenario: Medical diagnosis with 50 blood test features, need to identify which 5 tests to use in a rural clinic (tests are expensive).

Analysis:

- Need original features (specific blood tests)
- Cost constraint (can only afford 5 tests)
- Must be interpretable to doctors
- Have labeled training data (diagnoses)

Best approach:

1. Filter: Use correlation with diagnosis to narrow to 15-20 tests
2. Wrapper: Use forward selection with logistic regression to find best 5
3. Validate: Ensure selected tests are medically meaningful

Why not PCA? PCA would create combinations of tests, but clinic needs to know *which specific tests* to run.

9.2 Example 2: PCA Calculation

Given: 3 data points in 2D

x_1	x_2
1	2
2	3
3	5

Step 1: Center the data

- $\bar{x}_1 = (1 + 2 + 3)/3 = 2$
- $\bar{x}_2 = (2 + 3 + 5)/3 = 10/3 \approx 3.33$
- Centered: $\begin{bmatrix} -1 & -1.33 \\ 0 & -0.33 \\ 1 & 1.67 \end{bmatrix}$

Step 2: Covariance matrix

$$\Sigma = \frac{1}{n-1} X^T X = \frac{1}{2} \begin{bmatrix} 2 & 3.67 \\ 3.67 & 7.33 \end{bmatrix} = \begin{bmatrix} 1 & 1.83 \\ 1.83 & 3.67 \end{bmatrix} \quad (23)$$

Step 3: Find eigenvalues

$$\det(\Sigma - \lambda I) = 0 \implies (1 - \lambda)(3.67 - \lambda) - (1.83)^2 = 0 \quad (24)$$

Solving: $\lambda_1 \approx 4.47$, $\lambda_2 \approx 0.20$

Step 4: Variance explained

- PC1: $\frac{4.47}{4.47+0.20} = 95.7\%$
- PC2: $\frac{0.20}{4.47+0.20} = 4.3\%$

Conclusion: First PC captures 95.7% of variance, can reduce to 1D!

Attendance	Pass/Fail
High	P
High	P
High	P
Low	P
Low	F
Low	F
Low	F

9.3 Example 3: Information Gain Calculation

Scenario: Predict if a student will pass (P) or fail (F) based on attendance.

Step 1: Root entropy

- $p(P) = 4/7, p(F) = 3/7$
- $H(\text{Root}) = -\frac{4}{7} \log_2 \frac{4}{7} - \frac{3}{7} \log_2 \frac{3}{7} = 0.985$

Step 2: Conditional entropy

- High attendance: 3P, 0F $\rightarrow H = 0$
- Low attendance: 1P, 3F $\rightarrow H = -\frac{1}{4} \log_2 \frac{1}{4} - \frac{3}{4} \log_2 \frac{3}{4} = 0.811$

Step 3: Information Gain

$$IG = 0.985 - \left(\frac{3}{7} \times 0 + \frac{4}{7} \times 0.811 \right) = 0.985 - 0.463 = 0.522 \quad (25)$$

Interpretation: Attendance is quite informative (IG = 0.522 out of max 1.0).

10 Key Formulas Summary

10.1 PCA Formulas

$$\text{Centering: } \mathbf{X}_{\text{centered}} = \mathbf{X} - \boldsymbol{\mu} \quad (26)$$

$$\text{Covariance: } \boldsymbol{\Sigma} = \frac{1}{n-1} \mathbf{X}^T \mathbf{X} \quad (27)$$

$$\text{Eigendecomposition: } \boldsymbol{\Sigma} = \mathbf{P} \mathbf{D} \mathbf{P}^T \quad (28)$$

$$\text{Projection: } \mathbf{X}_{\text{PCA}} = \mathbf{X} \cdot \mathbf{P}_k \quad (29)$$

$$\text{Reconstruction: } \mathbf{X}_{\text{approx}} = \mathbf{X}_{\text{PCA}} \cdot \mathbf{P}_k^T \quad (30)$$

$$\text{Variance explained: } \frac{\sum_{i=1}^k \lambda_i}{\sum_{i=1}^m \lambda_i} \quad (31)$$

10.2 Filter Method Formulas

$$\text{Variance: } \text{Var}(X) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \quad (32)$$

$$\text{Pearson Correlation: } r_{XY} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \quad (33)$$

$$\text{Entropy: } H(X) = - \sum_i p(x_i) \log_2 p(x_i) \quad (34)$$

$$\text{Information Gain: } IG(S, A) = H(S) - \sum_v \frac{|S_v|}{|S|} H(S_v) \quad (35)$$

$$\text{Chi-Square: } \chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \quad (36)$$

$$\text{Expected Frequency: } E_{ij} = \frac{\text{Row}_i \times \text{Col}_j}{\text{Total}} \quad (37)$$

$$\text{Mutual Information: } I(X; Y) = H(X) + H(Y) - H(X, Y) \quad (38)$$

11 Common Exam Questions & Answers

11.1 Conceptual Questions

Q1: Why does PCA fail in high dimensions?

A: Trick question! PCA is specifically *designed* for high dimensions. However:

- PCA assumes linear relationships (fails with non-linear data)
- If data is uniformly distributed (no variance direction), PCA doesn't help
- Eigenvalue computation becomes numerically unstable for very high m

Q2: When would you prefer feature selection over PCA?

A:

1. Need interpretable original features (medical diagnosis)
2. Features are expensive to collect (want to know which to measure)
3. Categorical features (PCA requires numeric)
4. Regulatory requirements (must explain which features used)

Q3: Can you use PCA for classification?

A: Yes, but PCA is unsupervised (doesn't use labels). Better approach:

1. Use PCA to reduce dimensions
2. Then train classifier on reduced features
3. Alternative: Use LDA (Linear Discriminant Analysis) which is supervised

Q4: Why normalize features before PCA?

A: Features with larger scales dominate the principal components.

- Example: Height (150-200 cm) vs. Weight (50-100 kg) vs. Age (20-80 years)
- Without normalization, height variance dominates
- Normalization ensures equal importance

Q5: Difference between filter and wrapper methods?

A:

- **Filter:** Evaluate features using statistics (fast, independent of model)
- **Wrapper:** Evaluate features using model performance (slow, model-specific)
- **Analogy:** Filter is like pre-screening resumes, Wrapper is like interviewing candidates

11.2 Calculation Questions

Q6: Given covariance matrix, find variance explained by first PC

Given: $\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$, eigenvalues: $\lambda_1 = 5.24$, $\lambda_2 = 1.76$

Solution:

$$\text{Variance explained} = \frac{5.24}{5.24 + 1.76} = \frac{5.24}{7} = 0.749 = 74.9\% \quad (39)$$

Q7: Calculate information gain

See Example 3 in Section 9.3 for full worked solution.

Key steps:

1. Calculate root entropy
2. Calculate entropy for each subset after split
3. $IG = \text{Root entropy} - \text{Weighted average of subset entropies}$

12 Exam Tips & Common Mistakes

12.1 PCA Common Mistakes

1. **Forgetting to center data** → Always subtract mean!
2. **Not scaling when needed** → Check if features have different units
3. **Using all components** → Defeats the purpose of dimensionality reduction
4. **Thinking PCA is supervised** → PCA doesn't use labels
5. **Confusing eigenvectors and eigenvalues** → Values = variance, Vectors = directions

12.2 Feature Selection Common Mistakes

1. **Using correlation for categorical data** → Use chi-square or information gain
2. **Forgetting feature interactions** → Filter methods evaluate independently
3. **Overfitting with wrappers** → Use cross-validation
4. **Not considering computational cost** → Wrappers are expensive
5. **Ignoring domain knowledge** → Sometimes domain experts know best features

12.3 Exam Strategy

For conceptual questions:

- Always explain *why*, not just *what*
- Give examples to illustrate
- Compare/contrast when asked to choose

For calculation questions:

- Show all steps clearly
- Label intermediate results
- Check if answer makes sense (e.g., variance explained should be 0-1)
- Units matter (especially in PCA)

For method selection questions:

- Consider: dataset size, feature types, interpretability needs, computational budget
- Justify your choice with specific reasons
- Acknowledge trade-offs

13 Quick Reference Checklist

13.1 PCA Checklist

- ☐ Understand curse of dimensionality
- ☐ Know when to use PCA vs. feature selection
- ☐ Can explain maximizing variance = minimizing error
- ☐ Know all 5 steps of PCA algorithm
- ☐ Can calculate variance explained
- ☐ Understand scree plot and elbow method
- ☐ Know PCA applications
- ☐ Remember: PCA is unsupervised and linear

13.2 Feature Selection Checklist

- ☐ Understand combinatorial explosion (2^n subsets)
- ☐ Know difference: unsupervised vs. supervised
- ☐ Can calculate variance and correlation
- ☐ Understand entropy and information gain
- ☐ Can perform chi-square test
- ☐ Know filter vs. wrapper trade-offs
- ☐ Understand forward selection and backward elimination
- ☐ Know when to use which method

14 Final Summary

Dimensionality Reduction solves the curse of dimensionality through two main approaches:

14.1 Feature Extraction (PCA)

- Creates new features as linear combinations
- Unsupervised (doesn't use labels)
- Maximizes variance, minimizes reconstruction error
- Best for: visualization, compression, correlated numeric features
- Loses interpretability of original features

14.2 Feature Selection

- Selects subset of original features
- Can be unsupervised or supervised
- **Unsupervised:** Variance, correlation, mutual information
- **Filter (Supervised):** Fast, use statistics (correlation, info gain, chi-square)
- **Wrapper (Supervised):** Slow, use model performance (forward, backward, GA)
- Best for: interpretability, categorical features, expensive measurements

Remember: No one-size-fits-all solution. Choose based on:

- Dataset characteristics (size, type, dimensionality)
- Computational resources
- Interpretability requirements
- Domain knowledge
- Downstream task

Good luck on your exam!